

# Changyong Kwak

Madison, WI | (608) 515-0546 | [ckwak7@wisc.edu](mailto:ckwak7@wisc.edu) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## EDUCATION

### University of Wisconsin-Madison

*Bachelor of Science in Computer Science*

Expected December 2027

*Madison, WI*

## EXPERIENCE

### AI Research & Software Engineering Intern

Jun 2026 - Aug 2026

*Summer of AI Laboratory (SAIL) | N+1 Institute @ UW-Madison*

*Madison, WI*

- Engineered a Python agent-evaluation platform spanning **528 cases across 11 task families** by enforcing typed runtime contracts, fail-closed execution, scoped memory, and strict graders across **3 versioned domain packs**.
- Co-developed a browser-based Agent Lab that streams plans, model output, tool calls, and verifier verdicts over Server-Sent Events, giving engineers live inspection of workspace state and generated artifacts.
- Hardened concurrent writes and crash recovery with marker-last commits, writer locks, hash-bound manifests, exact resume, and corruption detection, verified by **461 native Windows ARM64 tests**.

### Software Engineer Intern

Feb 2026 - Jul 2026

*Contoro Robotics*

*Austin, TX*

- Shipped a full-stack control platform for **3 Isaac Sim robots** by building a React/TypeScript frontend on a **13-endpoint FastAPI backend**, giving **100+ users** browser access without local GPU or ROS setup.
- Orchestrated command sequencing across **4 production adapter modes** with per-robot queues, session locks, cancellation, timeout recovery, and fallback paths over a ROS 2 WebSocket bridge.
- Validated control-service reliability with **27 pytest cases** covering malformed input, cancellation, concurrency, and timeouts, then automated Python 3.10 to 3.12 tests, lint, type, and packaging gates in CI.

### Undergraduate Research Intern

May 2026 - Present

*CATS Lab @ UW-Madison*

*Madison, WI*

- Reproduced a **420K-parameter PyTorch Transformer** on **97,499 observations from 82 trajectories**, using 12 car-following features and trajectory-level 70/15/15 splits to prevent window leakage.
- Stress-tested generalization across **3 random seeds and 3 held-out recording folds**, measuring **0.834 mean trajectory AUROC** while documenting label/session confounding and validation-test gaps.

### Creator & Full-Stack Developer

Oct 2024 - Feb 2026

*GoViral | AI Video Automation Platform*

*Madison, WI*

- Launched a cloud-hosted video platform for **40+ beta users** and cut turnaround from **about 2 hours to under 10 minutes** by shipping Python/Flask APIs and parallel FFmpeg pipelines on Google Cloud Run.
- Instrumented API calls, latency, and pipeline failures in Google Cloud Monitoring across **40+ users**, isolating breaks in FFmpeg rendering, ElevenLabs voiceovers, and AssemblyAI captions during beta use.

## PROJECTS

### Husk | [GitHub](#) | TypeScript, Node.js, Docker, Vitest

Apr 2026 - Present

- Built an open-source npm/PyPI security gate reaching **97.3% precision and 82.8% F1 across 100 packages** by combining Babel AST analysis, seccomp-sandboxed Docker execution, and AI-assisted triage.

### LiveClaw | [GitHub](#) | Rust, Tauri, Python, macOS APIs

Mar 2026

- Designed a fail-safe desktop control loop across **3 control layers** by integrating a Rust/Tauri HUD, Python orchestration, and macOS input APIs behind **5 safety mechanisms** and 2-stage action verification.

### ClearMove | [GitHub](#) | Next.js, TypeScript, PostgreSQL, Jest

Nov 2025

- Created a Next.js/PostgreSQL rental-inspection application that placed **3rd of 92 teams** by converting leases, checklists, and timestamped photos into structured JSON records and evidence PDFs.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Rust, Java, SQL, C/C++, Go, Bash

**Software & Data:** React, Next.js, Tauri, FastAPI, Flask, Node.js, REST APIs, Server-Sent Events, PostgreSQL, Drizzle ORM, pandas, PyTorch, JSON serialization, data pipelines

**Systems & Cloud:** concurrency, distributed systems, ROS 2, NVIDIA Isaac Sim, Linux, AWS EC2/EKS, Kubernetes, Docker, Google Cloud Run/Monitoring, GitHub Actions CI/CD, Git, pytest, Vitest, Jest